

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.3604793053695368, median 0.31080498524070393, std: 0.26475621302361013
Gyroscope error (imu0): mean 4.355572912514156, median 2.2643224309183756, std: 8.121396457269444
Accelerometer error (imu0): mean 9.433663419695128, median 3.5555480651098, std: 20.9092784579921

Residuals

Reprojection error (cam0) [px]: mean 0.3604793053695368, median 0.31080498524070393, std:
0.26475621302361013
Gyroscope error (imu0) [rad/s]: mean 0.0024985416667021383, median 0.0012989115448493705, std:
0.004658778040885082
Accelerometer error (imu0) [m/s^2]: mean 0.06683882552671193, median 0.02519160338780982, std:
0.14814516403303432

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99965018 -0.00421749 0.02611009 0.0936423]
[0.02623108 0.03176994 -0.99915094 0.06549128]
[0.0033844 0.99948631 0.03186945 -0.05315445]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[0.99965018 0.02623108 0.0033844 -0.09514755]
[-0.00421749 0.03176994 0.99948631 0.05144143]
[0.02611009 -0.99915094 0.03186945 0.06468467]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.07695610735117799

Gravity vector in target coords: [m/s^2]
[0.02742874 1.54344115 -9.68428934]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [159.52102968304047, 158.38158812974802]

Principal point: [160.10793182038688, 160.5438343680662]

Distortion model: radtan

Distortion coefficients: [0.004139932795913384, -0.004483149069205775, 0.0010549037288716816, 0.004021189385902533]

Type: checkerboard

Rows

Count: 8

Distance: 0.02 [m]

Cols

Count: 11

Distance: 0.02 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

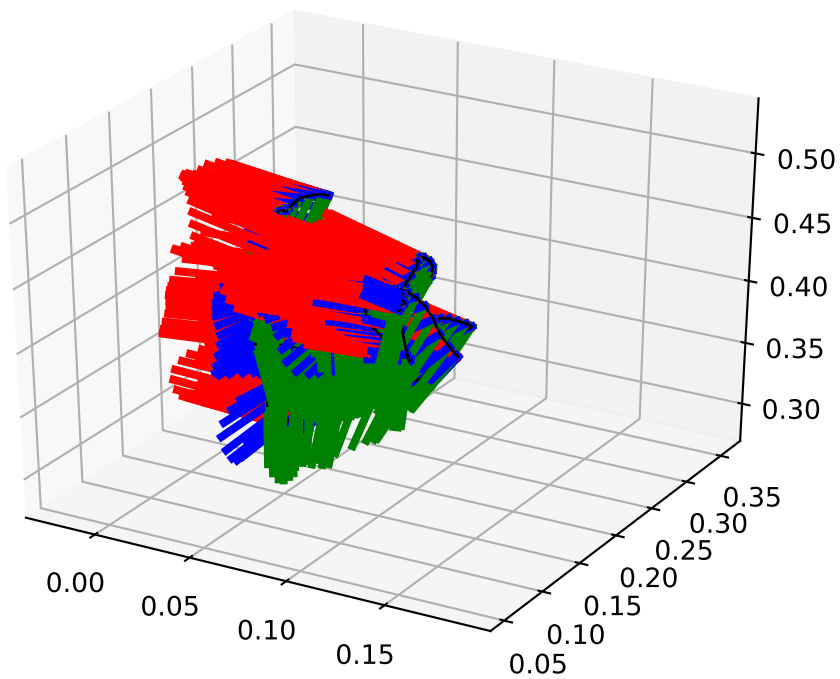
[0. 1. 0. 0.]

[0. 0. 1. 0.]

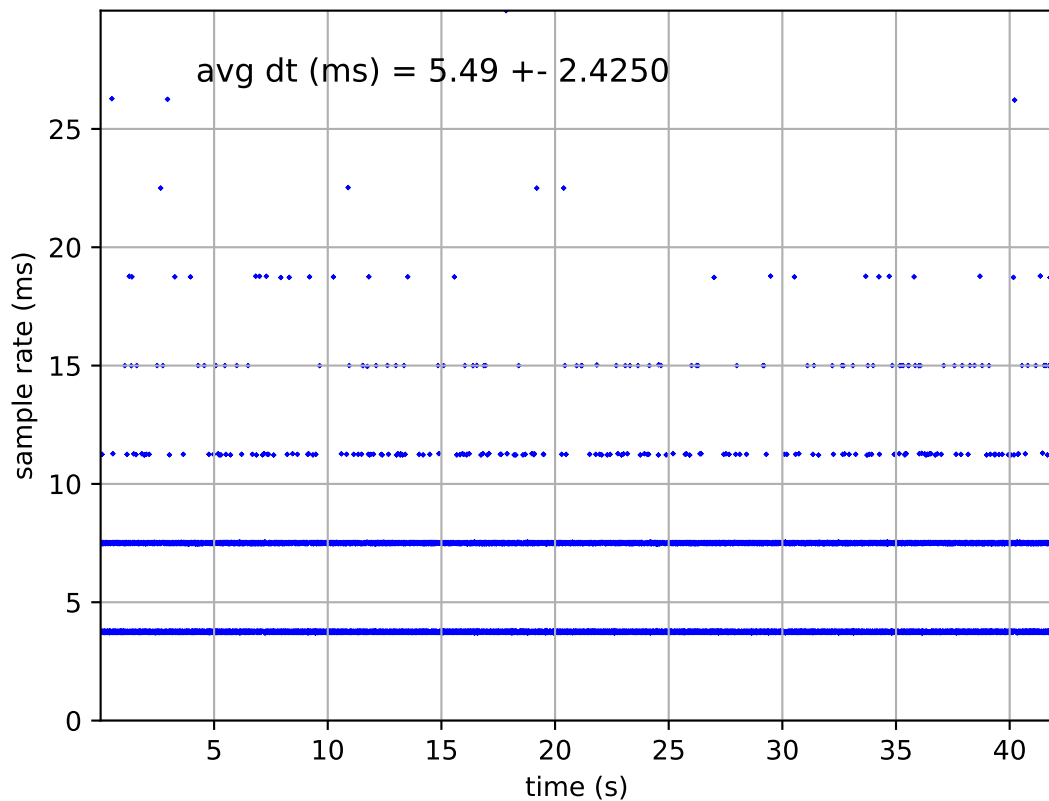
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

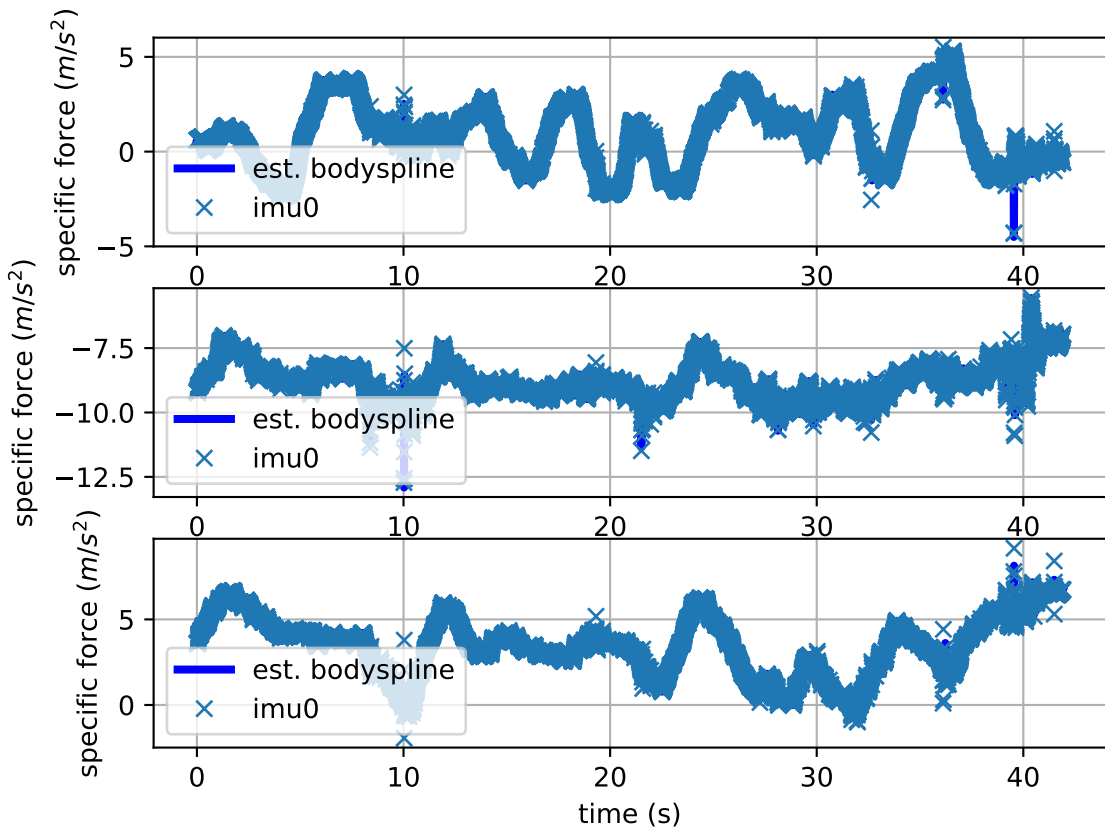
imu0: estimated poses



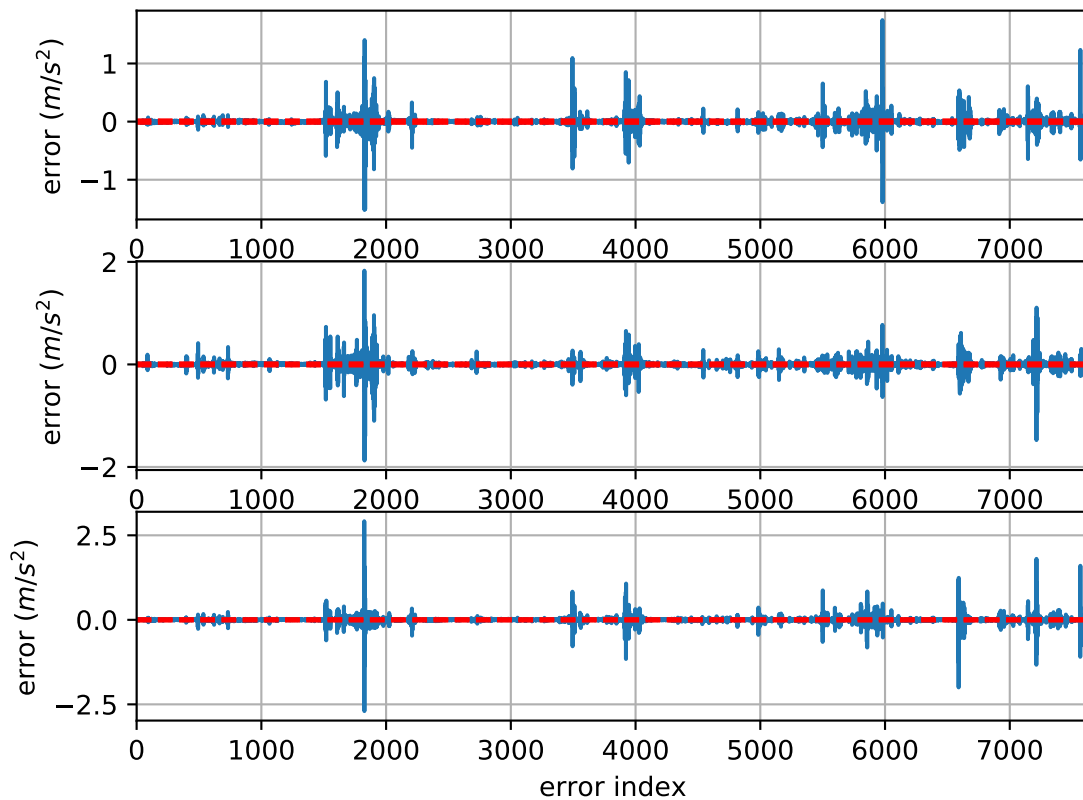
imu0: sample inertial rate



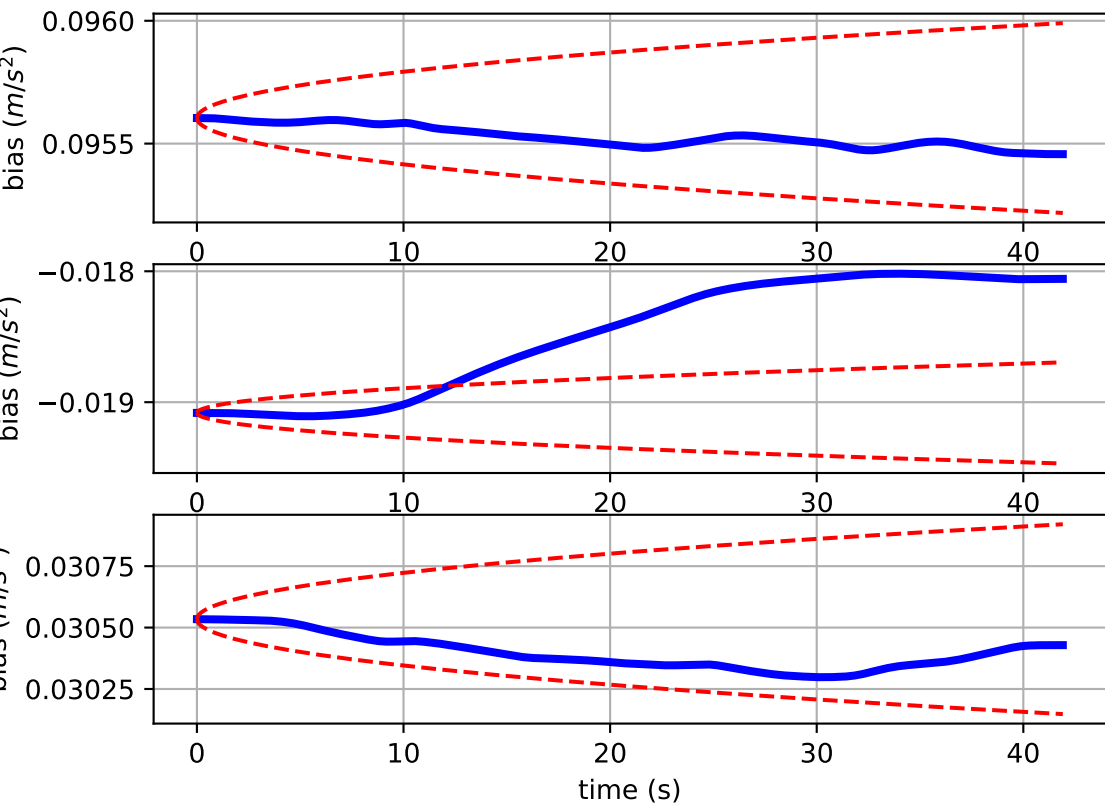
Comparison of predicted and measured specific force (imu0 frame)



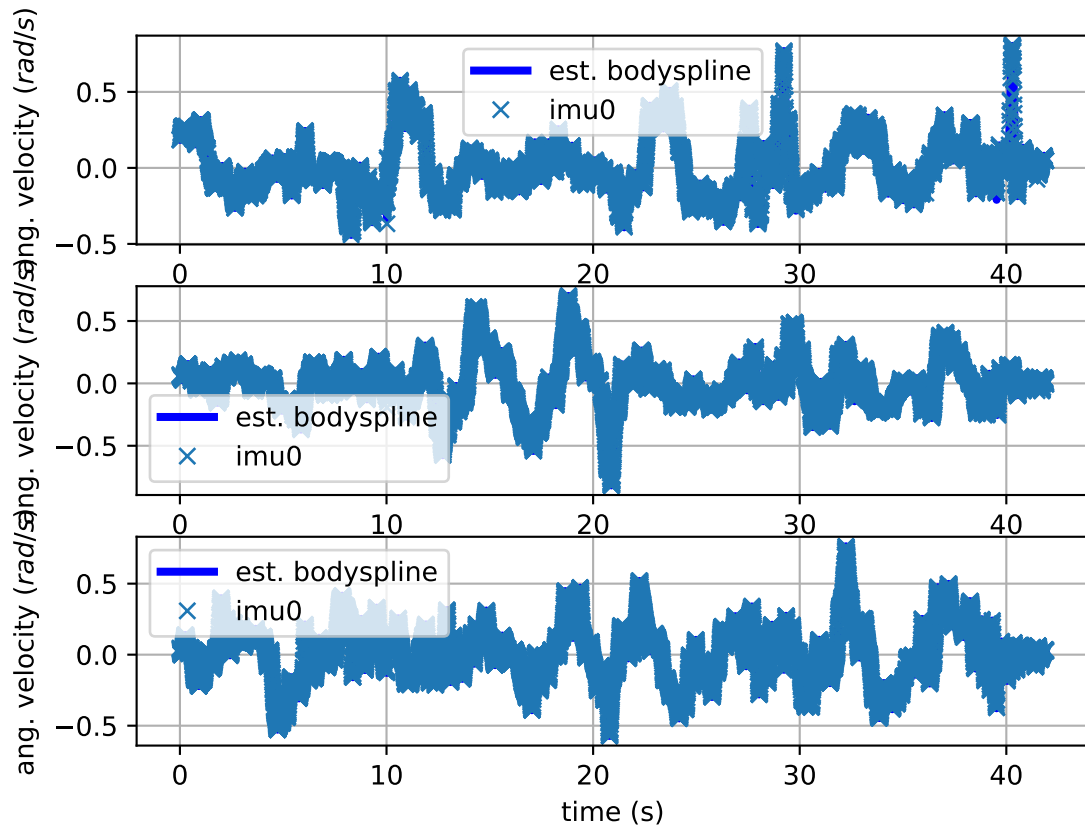
imu0: acceleration error



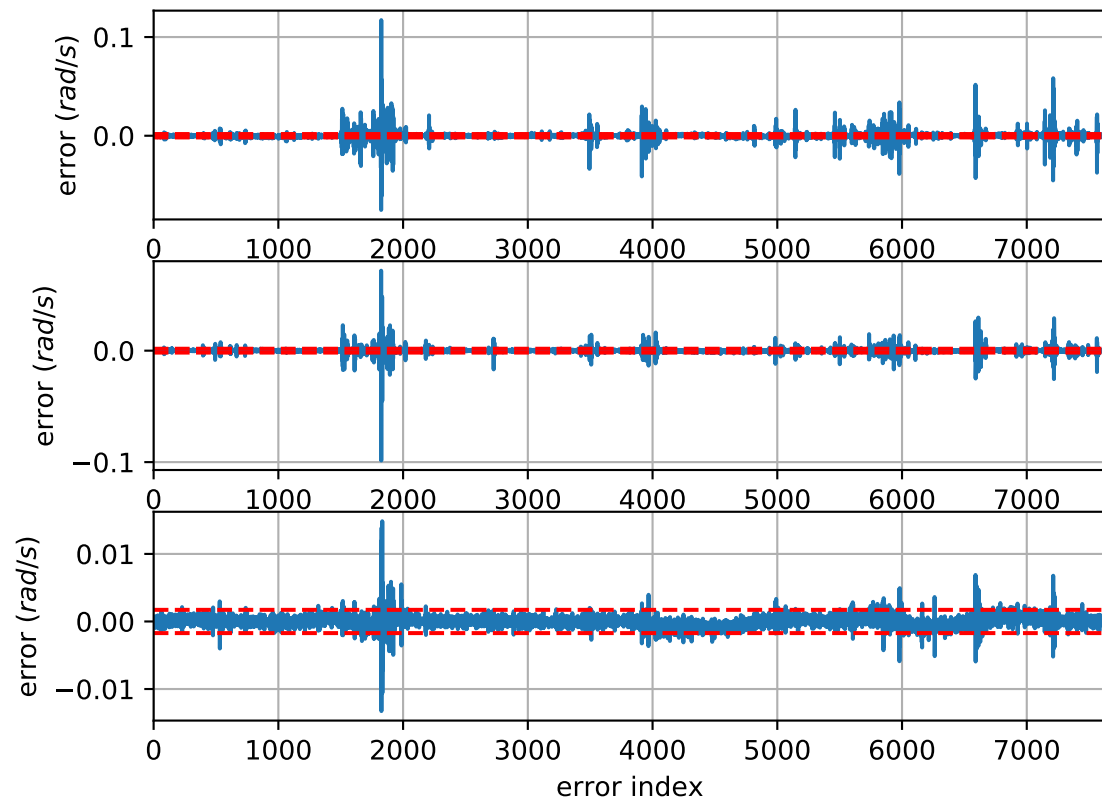
imu0: estimated accelerometer bias (imu frame)



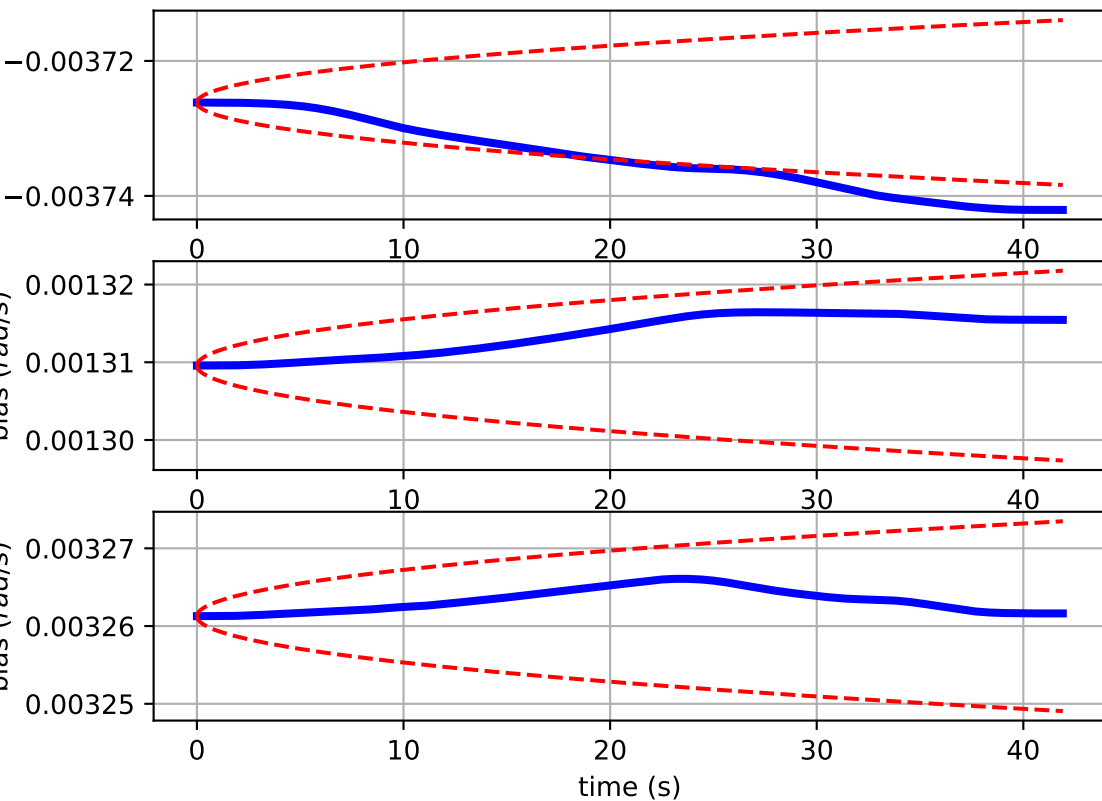
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

